

# Analysis of allele frequency changes among controlled *D. melanogaster* populations

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## ***Abstract***

Natural selection is the evolutionary mechanism by which the individuals most fit to survive and reproduce do just that—survive longer and reproduce more than other individuals within the same population. As a result, the allele frequencies of the population change by generation as the average phenotype shifts towards that which is most beneficial to the population. In the present experiment, we track the phenotypic expression of wing type among populations of the model organism, *D. melanogaster* (the fruit fly), to identify (a) whether evolution is occurring as a result of certain selective pressures, as well as (b) whether predictive modeling software can accurately describe the fluctuations of allele frequencies within these populations. We hypothesized that a population of fruit flies facing sexual selection as its primary selective pressure would evolve to exhibit an increase in the frequency of the allele (for wild-type wings) that confers greater reproductive success. Furthermore, we hypothesized that a population of fruit flies facing both sexual selection and natural selection (in the form of phenotype-specific predation) would evolve to increase the frequency of the same allele since, in this scenario, we figure sexual selection to be a greater selective pressure than natural selection. Lastly, we hypothesized that the phenotypic expression of sixth-generation flies from both populations would best fit predictive Simulation 1, which satisfies our previous hypotheses concerning allele frequency changes. Our experiment consists of two caged populations of fruit flies, one with and one without a predator. We complete weekly sampling of the populations to track allele frequency shifts and, after ten weeks, gather final allele frequencies to statistically compare these allele frequencies with expected final allele frequencies (determined by the Hardy-Weinberg Principle.) Our results indicate that evolution is occurring in both caged populations, with allele frequencies shifting to favor wild-type wings. The results also indicate that none of the predictive simulations produce final allele frequencies that are statistically similar to those of our caged populations. The failure of these predictive simulations to accurately represent the observed allele frequencies of our caged populations is indicative of the limitations of these computer-generated models. Further experimentation is necessary to determine the factors that contribute to reproductive success and, by extension, relative fitness and, thereby, allow for the production of more accurate predictive models for allele frequency dynamics

## ***Introduction***

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Is it possible to truly see evolution in action? No—not quite—but we can simulate evolution with model organisms, like *Drosophila melanogaster*, to observe the evolution of

a population in which evolution occurs more rapidly. *D. melanogaster* is a prime candidate for ecological and evolutionary research because these fruit flies reproduce quickly and are relatively simple to work with and maintain. Larvae typically hatch in one day, and the metamorphosis to adult fruit flies occurs in as little as two days (Fernández-Moreno et al. 2007). Therefore, our populations will undergo over twenty generations, allowing for the evolution of various traits to occur and contributing to increased fitness within the population.

Evolution occurs as organisms adapt to their environments. Adaptation occurs due to natural selection, and natural selection is the evolutionary mechanism that acts on individuals within the population, causing adaptation. When some major environmental change happens (e.g., natural disasters, bombings, urbanization, etc.), the individuals that are best suited to the new environment are more likely to survive longer and, therefore, have more reproductive success. Reproductive success is the component that determines both individual fitness as well as relative fitness, measures of how much of an individual's genotype is passed on to the next generations. By these means, organisms adapt and, with time (many generations), these adaptations contribute to new, distinct features or functions—in other words, new traits evolve (Walsh 2000). However, we do not yet understand the totality of the factors that contribute to reproductive success and, thereby, individual and relative fitness. This lack of knowledge contributes to the probable inaccuracy of mathematical models in predicting fitness and other related components that catalyze evolution.

In the present experiment, we seek to determine whether evolution occurs in two different *D. melanogaster* populations by tracking the frequency of the allele for wild-type wings ( $w_A$ ). The alternative, recessive trait ( $w_a$ ) confers what is referred to as vestigial wings, which are non-functional. Furthermore, sexual selection acts upon fruit fly populations and negatively affects vestigial-winged individuals, whose wings are deemed unsatisfactory by potential mates (Pezzoli et al. 1986). Because sexual selection differentially affects fruit flies based on wing type, we hypothesize that flies with vestigial wings will have lower reproductive success than flies with wild-type wings, causing the trait for wing type to evolve such that the frequency of allele  $w_A$  increases. In another cage, however, we introduce a predator that affects wild-type flies. Because of natural selection that occurs via predation, fruit flies with wild-type wings will be selected against, but since sexual selection also causes selection against vestigial flies (to a potentially greater extent), we hypothesize that the trait for wing type will evolve, resulting in increasing  $w_A$  frequency (Thomson et al. 2006). And with respect to predictive models, we hypothesize that Simulation 1 (see Figure 1) would best exemplify the generation six allele frequencies of our caged populations since the baseline relative fitness values of the simulation allow for the generation of reasonable allele frequency results that correlate with calculated expectations for allele frequency.

## ***Materials and Methods***

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We began the setup of the experiment by preparing two cages: one with and one without a “predator.” To control predatorial behavior in a way that detrimentally affected only fruit flies capable of flight, we inserted a small section of fly paper into Cage 2. The fly

paper, as our predator, captures and kills fruit flies that land on its surface. Vestigial flies, having damaged wings and being thereby incapable of flight, are not prone to predation by fly paper—natural selection acts upon the fruit fly population, negatively affecting wild-type flies and positively affecting vestigial flies. After setting the cages, we determined a fair ratio of vestigial to wild-type flies for both cages to be 4:1—each cage would contain eighty vestigial fruit flies and twenty wild-type fruit flies. To separate vestigial flies from wild-type flies, we used CO<sub>2</sub> gas to anesthetize the flies and feathers to carefully count each type. Afterward, we administered food—"fly food" with added water—to each cage and left the fruit fly populations to reproduce for the period of one week. Next, to form predictions, we utilized a computer modeling system to produce five alternative simulations, each providing a different potential outcome for our generation six fly subpopulations. We manipulated the initial relative fitness variables for the  $w_A$  and  $w_a$  alleles in each simulation to produce different potential final allele frequencies. For each cage, we selected one simulation to serve as a prediction of allele frequency trends.

We completed weekly checks of the fly population by sampling about one hundred flies per cage and recording the proportions of vestigial and wild-type flies. After weekly checks, we feed the populations again. Weekly checks (including the week of setup) were completed for ten weeks total. During week ten, we gathered and recorded data for our final samples, then used all data collected over the course of the experiment to determine whether evolution had occurred to the fly populations.

To validate our hypotheses—that evolution had occurred within our fruit fly populations—we calculated, by means of the Hardy-Weinberg principle, the theoretical number of flies of each wing type that should exist at the sixth generation if evolution *had* not occurred. We then compared these expected sub-population sizes to the actual generation six sub-population sizes with the Chi-square test ( $\alpha=0.05$ ). Likewise, we used the Chi-square test to test our predictions, comparing the actual sub-population sizes in each cage with the chosen simulated sub-population sizes.

## Results

**Table 1:** Observed versus expected population sizes of vestigial and wild-type fruit flies from Cage 1 (no predator). Chi-square analysis of the differences between observed and expected population sizes yields  $p < 0.05$ , thereby rejecting the null hypothesis that evolution is not occurring.

|                        | Expected population size | Observed population size |
|------------------------|--------------------------|--------------------------|
| Vestigial ( $w_{aa}$ ) | 312                      | 79                       |
| Wild-type ( $w_{AA}$ ) | 175                      | 408                      |
|                        | p-value:                 | $3.72 \times 10^{-255}$  |

**Table 2:** Observed versus model-predicted population sizes of vestigial and wild-type fruit flies from Cage 1 (no predator). The model used for comparison is Simulation 1 (see Figure 1). Chi-square analysis of the differences between observed and model-predicted population sizes yields  $p < 0.05$ , thereby rejecting the null hypothesis that there is no statistical difference between model-predicted and observed population sizes.

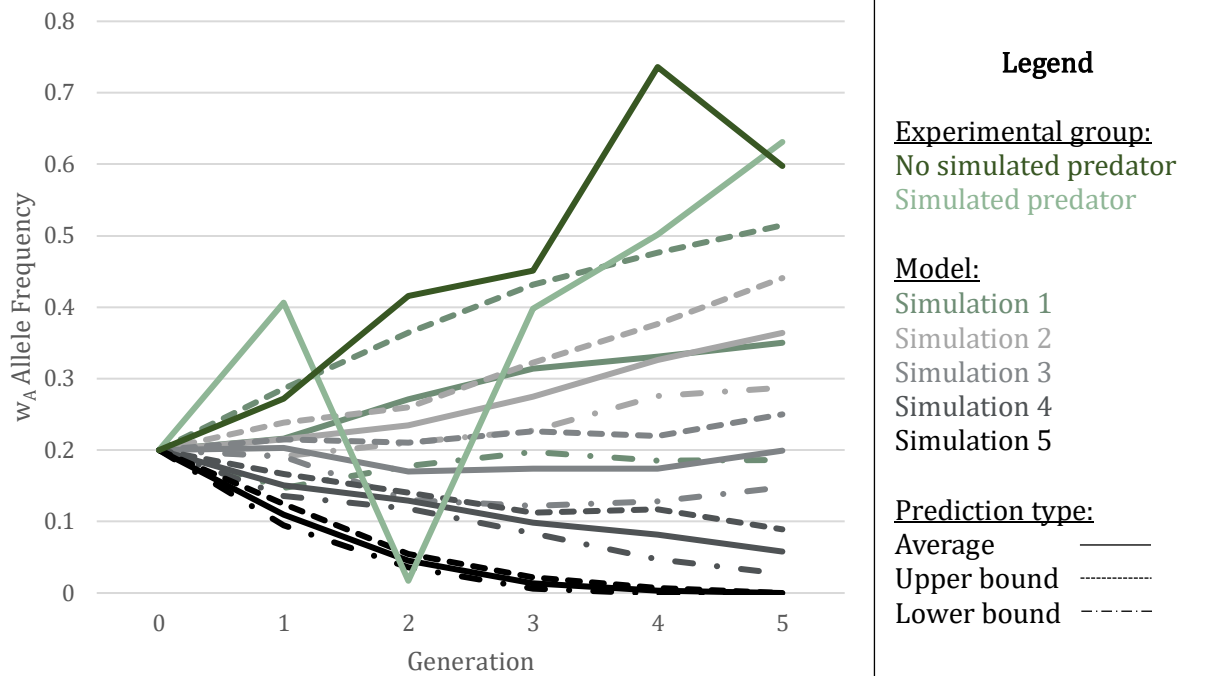
|                        | Model-predicted population size | Observed population size |
|------------------------|---------------------------------|--------------------------|
| Vestigial ( $w_{aa}$ ) | 206                             | 79                       |
| Wild-type ( $w_{AA}$ ) | 281                             | 408                      |
|                        | p-value:                        | $6.07 \times 10^{-107}$  |

**Table 3:** Observed versus expected population sizes of vestigial and wild-type fruit flies from Cage 2 (predator). Chi-square analysis of the differences between observed and expected population sizes yields  $p < 0.05$ , thereby rejecting the null hypothesis that evolution is not occurring.

|                        | Expected population size | Observed population size |
|------------------------|--------------------------|--------------------------|
| Vestigial ( $w_{aa}$ ) | 268                      | 57                       |
| Wild-type ( $w_{AA}$ ) | 151                      | 362                      |
|                        | p-value:                 | $1.88 \times 10^{-102}$  |

**Table 4:** Observed versus model-predicted population sizes of vestigial and wild-type fruit flies from Cage 2 (predator). The model used for comparison is Simulation 1 (see Figure 1). Chi-square analysis of the differences between observed and model-predicted population sizes yields  $p < 0.05$ , thereby rejecting the null hypothesis that there is no statistical difference between model-predicted and observed population sizes.

|                        | Model-predicted population size | Observed population size |
|------------------------|---------------------------------|--------------------------|
| Vestigial ( $w_{aa}$ ) | 177                             | 57                       |
| Wild-type ( $w_{AA}$ ) | 242                             | 362                      |
|                        | p-value:                        | $1.88 \times 10^{-32}$   |



**Figure 1:** Frequency of the  $w_A$  allele for wing type in two experimental *D. melanogaster* populations across six generations, compared to five different predictive models for  $w_A$  allele frequency. 95% confidence intervals are also displayed for each simulation (upper and lower bounds). For reference, we statistically compared the final phenotypic expression results for the caged populations to those of Simulation 1.

As shown in Tables 1 and 3, after six generations, wild-type fruit flies overwhelmingly outnumber vestigial flies, regardless of the presence of a predator. The experimental population sizes of vestigial and wild-type flies also deviate significantly ( $p < 0.05$ ) from expected values for these population sizes (see Tables 1 and 3). Compared to the expected model of best fit, Simulation 1, the experimental population sizes of vestigial and wild-type flies vary significantly ( $p < 0.05$ ) for both cages (see Tables 2 and 4). In Cage 1,  $w_A$  allele frequency increases over the span of six generations, and  $w_a$  allele frequency decreases equivalently (see Figure 1). In Cage 2,  $w_A$  allele frequency initially decreases but rebounds after three generations, leading to an overall increase in  $w_A$  allele frequency; alternatively, the frequency of the  $w_a$  allele in Cage 2 generally decreases across six generations (see Figure 1).

## Discussion

Initially, we predicted that evolution would occur within the fruit fly populations in Cages 1 and 2, and the frequency of the  $w_A$  allele would generally increase—the main driver of this frequency change being sexual selection. Although wild-type flies, specifically, in Cage 2 faced selective pressure through predation, we hypothesized that this selective pressure would not cause as great a reduction in the  $w_A$  allele since sexual selective pressure, based on research by Pezolli et al., would cause such a drastic reduction of the

frequency of the  $w_a$  allele (1986). Concerning the phenotypic expression data we collected, predictive data based on the Hardy-Weinberg Principle, and Chi-square analysis of the differences between experimental and predictive data, we reject the null hypotheses (with  $p < 0.05$  for both comparisons) that suggest evolution is not occurring within our caged populations (see Tables 1 and 3). Furthermore, we had predicted that Simulation 1 would best represent the six-generation change in allele  $w_A$  frequency for both Cages. Chi-square analysis of the difference between phenotypic expression of caged and modeled populations resulted in the rejection of the null hypothesis—that phenotypic expression differences between each cage and Simulation 1 predictions, respectively, are statistically irrelevant (see Tables 2 and 4). Therefore, our data has validated the hypotheses that evolution is occurring in both cages but has invalidated our hypotheses that suggest predictive Simulation 1 would accurately represent our raw data. In fact, the question of which simulation would accurately represent our data was farcical (though unintentionally), in that, none of the simulations accurately predicted the final phenotypic variation of our caged populations.

The failure of the computer modeling system to generate accurate simulations of allele frequency changes for theoretical populations points to the unpredictability of evolution. Though these simulations were grounded upon relative fitness values that our research group chose, we were incapable of predicting the true relative fitness values for alleles  $w_A$  and  $w_a$ . In a study of *D. melanogaster* population dynamics, the experimenter notes that relative fitness is a measure of the reproductive success (number of progeny) of an individual as compared to the average reproductive success of the population—however, in the present context, we consider relative fitness in terms of subpopulations (wild-type and vestigial) in comparison with the population as a whole (Barker 1963). Moreover, the computer modeling system is unable to account for additional factors that may cause relative fitness to change. Regarding the present experiment, to assume that these modeling systems may be capable of predicting allele frequency changes, we must also assume that wing type is the *only* genetic component driving differences in relative fitness. In reality, relative fitness is a constantly changing value, dependent on a multitude of genetic factors (Horn 1974). With the present experimental setup, we cannot control the influence of other factors (e.g., aggression) on the relative fitness of each fruit fly subpopulation. Even considering the true relative fitness of the alleles for wing type is challenging since *D. melanogaster* is too complex an organism—we cannot control other factors that may contribute to changing allele frequencies and, therefore, cannot postulate that changes in wing type allele frequencies are due entirely to the consequences of having a particular wing type.

Fortunately, for the caged populations of fruit flies, the fluctuation of relative fitness does not affect the legitimacy of our results. However, from a broader perspective, to produce more accurate predictive models for allele frequency dynamics, we must carry out experiments with organisms that do not reproduce so quickly and in such great quantity. The criterion that makes *D. melanogaster* a suitable model organism for the present experiment simultaneously renders this organism an unwise choice for reproduction-measuring experiments. Experimentation with organisms like mice, whose reproductive cycles allow for greater control and measurement accuracy regarding changes in relative fitness, may serve as better model organisms for understanding the factors that influence

relative fitness (Horn 1974). These experiments to determine the factors that influence fluctuations in relative fitness are important since the determination of such factors may allow for increased accuracy of predictive models for allele frequency changes across several generations. Ultimately, the purpose of these experiments is to understand how different types of selection may influence population dynamics as a whole. To better understand the interplay of these larger principles, however, we must also understand the correlation between natural selection, relative fitness, and the possibly infinite number of genetic and environmental factors that influence both.

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### ***Literature Cited***

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- Barker, J. S. F. 1963. The Estimation of Relative Fitness of *Drosophila* Populations. II. Experimental Evaluation of Factors Affecting Fitness. *Evolution* 17:56–71.
- Fernández-Moreno, M. A., C. L. Farr, L. S. Kaguni, and R. Garesse. 2007. *Drosophila melanogaster* as a Model System to Study Mitochondrial Biology. Pages 33–49 in D. Leister and J. M. Herrmann, editors. *Mitochondria*. Humana Press, Totowa, NJ.
- Horn, J. M. 1974. Aggression as a component of relative fitness in four inbred strains of mice. *Behavior Genetics* 4:373–381.
- Pezzoli, C., D. Laporta, G. Giorgi, D. Guerra, and S. Cavicchi. 1986. Fitness components in a vestigial mutant strain of *Drosophila melanogaster*. *Bolletino di zoologia* 53:351–354.
- Thomson, R. L., J. T. Forsman, M. Mönkkönen, M. Hukkanen, K. Koivula, S. Rytönen, and M. Orell. 2006. Predation risk effects on fitness related measures in a resident bird. *Oikos* 113:325–333.
- Walsh, D. M. 2000. Chasing shadows: natural selection and adaptation. *Studies in History and Philosophy of Science Part C: Studies in History and Philosophy of Biological and Biomedical Sciences* 31:135–153.